Zen Andrei Obrero

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Portfolio | GitHub

WORK EXPERIENCE

Uzaro Solutions Technology Inc.
Nov 2024 - Sept 2026

Technology Developer — *Quezon City*

- Developed production-grade frontend interfaces and reusable component libraries using React and TypeScript, establishing shared design patterns and type-safe contracts adopted across 8+ company-wide projects
- Designed backend data models (MongoDB/Prisma) that unified frontend-backend contracts, conducted code reviews on team PRs, and managed CI/CD pipelines via GitHub Actions with Firebase deployment - authoring unit and integration tests in Jest that reduced post-release bug reports by ~25%
- Deployed multiple backend services to Google Cloud Platform (Cloud Run)
- Participated directly in client-facing meetings across the full project lifecycle: project kickoff, requirements gathering, sprint planning, sprint review/demo, UI/UX review, progress reports, UAT review, go-live/deployment, and handover/training sessions
- Drove sprint delivery across 6+ concurrent platform builds (IoT, LMS, HMS, HRIS, Inventory, Order Management) in an Agile environment using GitHub Projects and a custom-built PMS, accelerating delivery velocity through AI-assisted prototyping and scaffolding with Claude Opus 5, Codex, Grok Build, and Gemini 3.1 Pro
- Integrated smart door locks, switch/outlet relays, and IoT gateways via REST APIs into a centralized React dashboard, enabling real-time device control and monitoring for 50+ connected endpoints with sub-500ms response latency
- **HRIS (Bandai Namco)** - Built and shipped admin, HR, and employee surfaces covering employee lifecycle, payroll/payslips, attendance with Hikvision biometric device integration, leave/benefits, schedules, and BIR reporting using React/TypeScript and a Node/Prisma API, with multi-environment delivery via GitOps (DEV/UAT/PROD)

IPSolutions Inc.
Mar 2024 - May 2024

IT Intern — *Pasig City*

- Designed and delivered 3 responsive client-facing landing pages including the company's main website using HTML/CSS and Bootstrap, each passing cross-browser QA on first review within the 3-month internship
- Collaborated with the development team to implement features following SDLC best practices, gaining foundational experience in code versioning with Git and team-based delivery workflows

PROJECT EXPERIENCE

Shared Stack: *React, TypeScript, Node.js, Express, MongoDB, Prisma ORM, Zod, TanStack Query, Zustand, JWT, Tailwind CSS*

ERP Platform Suite (Uzaro Solutions)

Architected and developed interconnected ERP modules as part of a multi-tenant enterprise platform:

- **HRIS** - Built for Bandai Namco, a manufacturing company, covering employee lifecycle, payroll/payslips, attendance and biometric device integration, leave/benefits, schedules, and BIR reporting for multi-tenant admin/HR/employee surfaces - the most data-intensive module, handling 6,000+ employee records with role-scoped visibility across tenants
- **Product and Assembly** - Built a production planning tool for Bandai Namco, a manufacturing company, coordinating product and assembly workflows - production schedules, assembly line planning, and operational visibility for manufacturing teams; React/TypeScript interfaces and backend data models for planning decisions, status tracking, and cross-team coordination
- **Inventory Management (GS1-Standard)** - Developed a GS1-compliant inventory system with standardized barcode/SKU identification, real-time stock tracking, and low-stock alerting - reducing manual stock audit time and enabling automated procurement triggers
- **Order Management / E-Commerce** - Built the employee partnership e-commerce platform handling product catalogs, cart workflows, and order lifecycle tracking; the most user-facing module with buyer, vendor, and admin role separation
- Additional modules: Project Management System, Supply Chain Management (integrated with Inventory and Order systems for end-to-end procurement-to-fulfillment)

Peck (Job Tracker & AI Resume Builder)

- Designed and built a mobile-first PWA job tracker with an AI resume builder for Philippine job seekers (React 19, Vite, TanStack Router/Query, Hono, Drizzle ORM, Zod, PostgreSQL on Supabase with row-level security, Typst); tracking is free and AI features are paid per use through a credit ledger
- Built a facts-only AI resume pipeline: an LLM fills fixed slots from user-confirmed facts, code assembles the document, and a custom validator rejects any line not traceable to a stored fact ID - it caught 1,630 of 1,630 injected mutants in mutation testing
- Implemented an append-only credit ledger with spend, refund and refill in single transactions, automatic refunds for failed AI jobs, plus AI cost controls (spend breaker, kill switch), an SSRF-safe job-post fetcher and per-user envelope encryption of stored personal data
- Produced ATS-checked one-page PDF resumes verified by two independent text extractors, and tested with Vitest and Playwright suites with axe accessibility checks, CI-verified row-level security and a 170 KB bundle budget

ALMA (Alternative Learning Management App)

- Architected a multi-tenant LMS serving 3 organizational tenants with role-based access for admins, instructors, and 200+ student accounts, implementing tenant-isolated data schemas and JWT-based authentication
- Built an assessment engine with automatic quiz scoring, grade computation, and analytics dashboards tracking learner performance across courses - reducing manual grading workload for instructors
- Developed attendance tracking and gradebook modules used by 15+ instructors across 40+ classes, enabling real-time class record management and automated academic report generation per tenant

HMS Reservation System

- Built a multi-tenant hotel management system enabling organizations to manage rooms, pricing, users, and reservations with tenant-isolated data architecture
- Designed and implemented core booking functionality including room type configuration, dynamic pricing rules, and full reservation lifecycle management from search to checkout

EDUCATION

Biliran Province State University Naval, Biliran

Bachelor of Science in Computer Science | With Honors *Graduated May 2024*

- Algolympics Champion - 1st Place, campus-wide algorithm programming contest (2nd Year)
- Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Engineering, Object-Oriented Programming

TECHNICAL SKILLS

Frontend: React, TypeScript, Next.js, Tailwind CSS, Bootstrap, HTML/CSS, Zustand, TanStack Query, Socket.io

Backend: Node.js, Express, Python (FastAPI), REST API Design, JWT Authentication, Zod

Databases & ORM: MongoDB, PostgreSQL, MySQL, Prisma ORM

Testing: Jest, pytest, Playwright

DevOps & Deployment: Git, GitHub, GitHub Actions (CI/CD), Google Cloud Platform (Cloud Run), Firebase Hosting, Docker

Languages: TypeScript, JavaScript, Python, Java, C++, PHP

Project Management: GitHub Projects, Custom PMS Tool, Agile/Scrum

AI-Augmented Workflow: Claude Opus 5, Codex, Grok Build, Google Stitch / Gemini 3.1 Pro

Domain Experience: IoT Smart Devices, Education/LMS, Hotel Management, Manufacturing / Production Planning, ERP (HRIS, Inventory/GS1, Order Management, Supply Chain, Project Management)